

GROUND LADDERS

Job Performance Requirements

NFPA 1001

4.3.6*

4.3.9*

4.3.10*

4.5.1

Note: All objectives in this section are testable.

Introduction

Subject: Ground Ladders

Note: ALL ladders must be verified verbalizing either safe **OR** secure to climb.

1. When deploying a ladder, the scene shall be surveyed for any terrain or overhead obstructions (must be verbalized).
2. Students shall always remain in contact with the ladder while:
 - a. Deploying
 - b. Raising
 - c. Extending the Fly
 - d. Until properly relieved and heeled by other personnel
 - e. Until secured by an approved method
3. **While keeping your back straight, lifting and lowering techniques shall be performed with your legs, not your arms and back.**
4. When carrying a ladder, the Butt shall be facing forward in the direction of travel and slightly tilted down, so your vision is NOT obstructed.
5. When carrying tools up a ladder, both hands slide along the beams and shall remain in contact with the beams, NOT the rungs. **While climbing, the free hand shall slide up the beam, ensuring NOT to remove the hand and re-grasp the beam.**
6. When assisting a conscious victim down a ladder, both hands shall remain in contact with the back of the beams, NOT the rungs. **While descending, the hands shall slide down the beams, ensuring NOT to remove hands and re-grasp the beams.**
7. When removing a leg lock, student shall place both feet on the same rung (**neutral position**) before descending the ladder.
8. **After instructions are given to perform a skill, time begins when the student touches any tools, equipment, or hose to execute the required skill. Time ends when the student descends the ladder, is standing on the ground and informs the examiner skill/task is complete.**

Performance Objective

Subject: Ground Ladders

Task: Deploy a 24' Extension Ladder (One Firefighter Raise)

Note: Proper body mechanics shall be used when lifting and lowering ladders.

Procedure:

1. Confirm order with command.
2. Disengage the locking devices securing the ladder on the apparatus.

Note: Some apparatuses do not have ladders secured on the outside and are contained in a compartment. These ladders are deployed by sliding them out of the compartment from the rear of the apparatus.

3. Remove the roof ladder and place it on the ground, ensuring it is not directly exposed to the exhaust of the apparatus.
4. Remove the extension ladder, utilizing a low shoulder carry and with the bed section against your body. Carry the ladder with the butt end forward and carry slightly lowered.
5. Verify and verbalize any overhead or ground obstructions in the area.
6. Carry the ladder to the desired location, kneel and remove the ladder from your shoulder.
7. Place the ladder on its beam and lay the ladder flat, with the bed section down on the ground (fly section up).
8. Position yourself at the tip of the ladder, slide the butt of the ladder into the building, and raise the ladder to a vertical position with the fly section against the building.

Note: When carrying the ladder, you may “stick/spike” the butt into the building and directly raise the ladder while standing. Replaces steps 6-8.

9. Reposition the butt a safe distance (approximately 12-24 inches) from the building. This allows the ladder to be pushed into the building if individual is losing control.
10. Grasp the halyard and extend the fly section, without touching the building. Ensure both dogs are locked and press the tips into the building.
11. Grasp the halyard slack in your hand and lift the ladder while leaving the tips on the building for stability and reposition the butt approximately six (6) feet from the building. **This shall be done without walking backwards.**

12. Maintain contact with the ladder with one foot or tips of boots, then pass the excess halyard through the ladder (approximately waist high) between the rungs.
13. Grasp the excess halyard underneath the rung it was passed through and form a bight. Pull the slack out of the halyard and keep it taut.
14. Tie a Clove-Hitch on a bight around the upper rung the halyard was passed through, with a safety.
15. Rotate the ladder so the fly section is positioned out.

Note: Metal and fiberglass extension ladders shall be positioned with the fly section “out” for climbing. Wooden extension ladders shall be positioned with the fly section “in” for climbing.

16. After rotating the ladder, a proper climbing angle will be achieved and verified by utilizing the following methods:
 - a. The toes of both feet should touch the butt;
 - b. Stand erect with arms extending straight out and at shoulder level;
 - c. The hands should be positioned between the beams as if grasping a rung;
 - d. Maintain this position without leaning to compensate an incorrect angle.
17. Before climbing any ladder, verify and verbalize the following:
 - a. Four points of contact;
 - b. Both dogs are locked;
 - c. Halyard is tied and secured;
 - d. Proper climbing angle;
 - e. Ladder is safe **OR** secure to climb.
18. After verifying the ladder, heel the ladder by positioning yourself using one of the following methods:
 - a. Underneath: Stand with feet offset for stability, grasp both beams at eye level below a rung, apply constant pressure against the building while looking straight ahead.
 - b. Outside: Stand with one foot placed on the bottom rung, grasp both beams and press the ladder against the building.
19. While heeling the ladder, remain alert for falling objects and descending personnel.
20. Continue heeling the ladder until properly relieved.

Performance Objective

Subject: Ground Ladders

Task: Climb a Ladder

Procedure:

1. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
2. Grasp the rungs with the palms down and the thumbs underneath the rungs or hands on back of beams and slide hands on beams while ascending.
3. Arms shall be extended straight out. Ensure not to reach upward and pull with the arms. This will keep the body away from the ladder, allowing the knees to move freely.
4. While ascending the ladder, grasp and step on alternate rungs simultaneously, so both hands and feet do not occupy the same rung at any time.
5. Climb using your legs, do NOT use your arms.
6. Keep your eyes straight ahead with an occasional glance toward the tip of the ladder.
7. Climb the ladder smoothly, so there is a minimum amount of bounce and movement in the ladder.

Performance Objective

Subject: Ground Ladders

Task: Leg Lock on a Ladder

Procedure:

1. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
2. Climb the ladder and stop at the designated height where work is to be performed.
3. Step up one (1) rung and slide the opposite leg from the working side over the next rung (Up 1, Over 2).
4. Step down one (1) rung. With the leg wrapped around the locking rung, hook the foot on the beam (NOT the rung).
5. You should be standing on the rung initially stopped to perform work, with the opposite leg locked in place.
6. When removing a leg lock, candidates shall place both feet on the same rung (**neutral position**) before descending the ladder.

Performance Objective

Subject: Ground Ladders

Task: Lock on a Ladder with a Harness or Truck Belt

Note: Instructional purposes only for individuals who are physically unable to perform a leg lock. Training Centers are NOT required to purchase a Truck Belt. These circumstances will be considered and understood for the state exam.

Procedure:

1. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
2. The harness shall be fastened snug around the waist.
3. The locking device should be positioned to the side while ascending the ladder.
4. Climb the ladder and stop at the designated height where work is to be performed.
5. Reposition the locking device to the front and center.
6. Attach the locking device to the rung nearest the waist.

Performance Objective

Subject: Ground Ladders

Task: Climb a Ladder Carrying a Tool

Procedure:

1. Confirm order with command.
2. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
3. Grasp the tool needed to perform the work.
4. Place the free hand on the backside of the beam.
5. **While ascending the ladder, slide the free hand on the backside of the beam maintaining continuous contact with the beam, ensuring NOT to remove the hand and re-grasp the beam.**
6. If possible, the hand carrying the tool is slid along the opposite beam.
7. Climb the ladder and stop at the designated height where work is to be performed.
8. Make a leg lock on the opposite side work is to be performed or use a harness, as previously instructed.
9. **Alert personnel below of work to be performed.**
10. Remove leg lock and place feet in the **neutral position**.
11. Descend the ladder until standing on the ground.

Performance Objective

Subject: Ground Ladders

Task: Climb a Ladder with an Uncharged Hose Line

Procedure:

1. Confirm order with command.
2. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
3. Flake hose on the ground near the base of the ladder.
4. Facing the ladder, bring the hose up under the arm on the side work is to be performed.
5. Lay the hose across your chest and over the opposite shoulder, resting the nozzle at the small part of your back.
8. Ascend the ladder **grasping the rungs** or hands on back of beams and slide hands on beams while ascending.
6. until reaching the designated height where work is to be performed.
7. Apply a leg lock on the opposite side work is to be performed or use a harness, as previously instructed.
8. Bring the nozzle over the shoulder in front of you, ensure the bale is closed, and signal for water to charge the hose line.
- 9. Alert personnel below of work to be performed.**
10. Direct the nozzle down and away from the fire, then bleed the hose line and set the desired fog stream pattern if necessary.

Note: Do NOT direct the nozzle across the body to the opposite side work is being performed. This defeats the purpose of having a leg lock on the opposite side.

11. Remove leg lock and place feet in the **neutral position**.
12. Descend the ladder until standing on the ground.

Performance Objective

Subject: Ground Ladders

Task: Assist a Conscious Victim down a Ladder

Procedure:

1. Confirm order with command.
2. Verbally contact the victim, identify yourself, and inform them of the procedure about to occur.
3. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
4. **Ascend the ladder grasping the backside of the beams with both hands or hands on back of beams and slide hands on beams while ascending and stop on the rung directly below the victim.**
5. Maintain continuous contact with the beams throughout the entire process, **ensuring NOT to remove the hand and re-grasp the beam.**
6. Before descending the ladder, notify the process of stepping down.
7. **Descend the ladder only one (1) rung each time, bringing both feet to the neutral position. (Lower your right foot 1 rung, then lower your left foot even with the right).**
8. Do NOT alternate your foot pattern, begin each step with the same foot.
9. Continuously assure the victim throughout the process.
10. Repeat this process until reaching the ground with the victim.

Performance Objective

Subject: Ground Ladders

Task: Deploy a Roof Ladder (Multi-story Pitched Roof)

Procedure:

1. Confirm order with command.
2. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
3. Carry the Roof Ladder, in a low-shoulder carry, with the butt forward.

Note: The Roof Ladder can be carried with the tips forward.

4. Place the Roof Ladder on the ground at the base of the ladder to be climbed.
5. Open the hooks and ensure they're facing outward (away from you).
6. Raise the Roof Ladder and hook it to a rung on the ladder to be climbed.
7. Ascend the ladder grasping the rungs until your shoulder is approximately $\frac{2}{3}$ the height of the Roof Ladder ($\frac{1}{3}$ from the tips).
8. **Alert personnel below of work to be performed.**
9. Insert your hand through the rungs of the Roof Ladder and place it on your shoulder, **ensuring the hooks are facing outward.**
10. When the Roof Ladder is placed on the shoulder, ascend the ladder grasping the rungs with both hands or hands on back of beams and slide hands on beams while ascending.
11. Stop at the designated height where work is to be performed.
12. Make a leg lock on the opposite side work is to be performed or use a harness, as previously instructed.
13. Remove the Roof Ladder from your shoulder, place it on the roof, and slide it on its beam to the peak of the roof.
14. When the hooks are beyond the peak, lay the Roof Ladder flat with hooks down, and pull back to catch the hooks on the roof.
15. Remove leg lock and place feet in the **neutral position.**
16. Descend the ladder until standing on the ground.

Performance Objective

Subject: Ground Ladders

Task: Deploy a Roof Ladder on a Single Story or Low Roof Line (Pitched Roof)

Procedure:

1. Confirm order with command.
2. Verify by verbalizing either safe **OR** secure for climbing and heeled/secured to the building.
3. Carry the Roof Ladder, in a low-shoulder carry, with the butt forward.

Note: The Roof Ladder can be carried with the tips forward.

4. Place the Roof Ladder on the ground at the base of the ladder to be climbed.
5. Open the hooks and ensure they're facing outward (away from you).
6. Raise the Roof Ladder and place it parallel/adjacent to the ascending ladder, **ensuring the hooks are facing toward the roof.**
7. Ascend the ladder grasping the rungs or hands on back of beams and slide hands on beams while ascending.
8. Stop at the designated height where work is to be performed.
9. Make a leg lock on the opposite side work is to be performed or use a harness, as previously instructed.
10. **Alert personnel below of work to be performed.**
11. Grasp the farthest beam of the Roof Ladder and lift/pull it toward you, ensuring the hooks are facing outward. This will beam the Roof Ladder on the roof line.
12. Place it on the roof and slide it on its beam to the peak of the roof.
13. When the hooks are beyond the peak, lay the Roof Ladder flat with hooks down, and pull back to catch the hooks on the roof.
14. Remove leg lock and place feet in the **neutral position.**
15. Descend the ladder until standing on the ground.